

SYNTHESIS AND CHARACTERIZATION OF OXIDATIVELY DEGRADABLE POLYUREA BEARING DIACYLHYDRAZINE MOIETY IN THE MAIN CHAIN

Polyureas were successfully synthesized at room temperature from 4-aminobenzohydrazide and hexamethylene diisocyanate using N, N-dimethylformamide (DMF) or N-Methyl-2-pyrrolidone (NMP) as solvents through step-growth polymerization. The polymers contain diacylhydrazine and urea moieties in every repeating unit. Upon treatment with a calcium hypochlorite solution, these polymers were oxidized and degraded to the corresponding dicarboxylic acid, accompanied by the evolution of N₂ gas bubbles. The synthesized polyureas were also thermally stable and did not react with atmospheric oxygen. The polymerization processes were confirmed by the IR and NMR spectra. Moreover, polyureas had an extraordinary thin film-forming ability, which was also oxidatively degradable. Because of the formation of hydrogen bonding between polymer chains, it was difficult to surmount the energy barrier to dissolve the synthesized polymer. As a result, the synthesized polymer was found to be insoluble in most of the common solvents except NMP, DMF, and DMSO. Because of the properties of oxidative degradability, thermal stability, and film-forming ability, this type of polymer can be used for different purposes and can gather interest in scientific research and industrial manufacturing through more improvisations.